

How Many Metastable States Did You Actually Find?

Sampling Budget as a Confound in Dimensionality-Reduction Analyses of Molecular Simulation

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Abstract

A common claim in molecular simulation is that projecting a trajectory reveals N metastable states. Here N is operationally the number of Gaussian-mixture components selected in a low-dimensional embedding; without a kinetic model it should be read as an apparent-basin count, not a validated metastable-state count. Across two potentials with basin counts fixed by construction and two independent 100 ns alanine-dipeptide simulations, the BIC-selected component count rises strongly with sampling budget (Spearman $\rho = 0.87$ - 0.97 across systems and methods) although the reference landscape is unchanged. The trend persists under uniform thinning of a common reference trajectory, where coverage is approximately matched, and under IID Boltzmann sampling in the exact synthetic latent coordinates, which removes time correlation and projection learning. Other criteria exhibit different failure modes: AIC is frequently ceiling-censored or non-monotone, ICL never reaches the tested ceiling but remains method-dependently inaccurate, and silhouette and the simple elbow diagnostic are largely insensitive. BIC-selected recovery often peaks at an intermediate budget and then declines even while recovery at the reference component count is flat or improves. One exception is explicit: alanine TICA seed 1 shows no decline within the tested range because its high-recovery crossover occurs at the last budget. We recommend reporting sampling-budget curves, treating mixture components as candidate states until kinetically validated, and using ICL only as a less-ceiling-prone sensitivity analysis rather than a general default.

1 Introduction

Dimensionality reduction is the standard first step in characterizing the conformational landscape of a molecular system from simulation data. A trajectory of hundreds or thousands of coordinates is projected onto two or three dimensions using PCA, TICA, or increasingly a learned nonlinear embedding such as a VAE, and the resulting low-dimensional density is read for metastable states—the basins a molecule spends most of its time in and transitions between only rarely.

The sentence that reports this reading, “our analysis reveals N metastable states,” attributes N to the molecule. This paper asks how much of N is instead attributable to the sampling

budget: the number of frames the analyst happened to save.

The question matters because sampling budget is rarely fixed by the science. It is set by wall-clock time, hardware access, and project deadlines, and it varies by orders of magnitude across published studies of the same or similar systems. If reported state counts are sensitive to this variable, then a literature comparing state counts across studies—or a single study reporting a state count without its sampling context—is implicitly comparing sampling budgets as much as it is comparing molecules.

We test this directly using systems where the number of metastable states is known independently of any dimensionality reduction or clustering method: two synthetic potentials with basin count fixed by construction, and a real all-atom simulation of alanine dipeptide, whose three conformational states are defined by backbone dihedral angles rather than by any method under audit. On every system tested, the BIC-selected Gaussian-component count increases with sampling budget at fixed reference labels. A coverage-matched control shows that incomplete exploration is not a sufficient explanation.

This paper makes three contributions.

A controlled audit of a previously undocumented confound. We show that BIC-selected mixture-component count is a function of sampling budget in the audited pipelines, holding the reference landscape fixed by construction. The effect appears in two synthetic potentials of different dimensionality and basin count, and reproduces in two independent 100 ns all-atom simulations of a real biomolecule.

A control design that isolates the mechanism. By comparing a genuinely short trajectory against a uniform subsample of one long reference trajectory at matched sample size, we separate two candidate explanations that prior comparative work does not distinguish: whether the effect reflects incomplete exploration of the landscape, or the state-count estimator’s response to sample size independent of exploration. The effect persists under coverage matching on every system tested, which makes incomplete exploration insufficient as a sole explanation.

A practical recommendation, not just a diagnosis. Five selection diagnostics are compared on identical data. Their behavior is criterion dependent: BIC trends strongly with budget, AIC is often censored or non-monotone, and the integrated completed likelihood (ICL) never reaches the search ceiling on

either synthetic system, though it is not thereby accurate. We further localize where the resulting error occurs: across the seed-0 summary of nine method-system pairs, BIC-selected recovery declines from its peak to the largest budget tested. PCA and VAE decline sharply (43-84%). TICA’s smaller and more variable declines are associated with how much of the tested range remains after oracle-k recovery first exceeds 0.80; the second alanine seed shifts that crossover and shows no decline within range. This association is descriptive, not a general mechanism established from three systems.

We do not claim that any published metastable-state count is wrong. The claim is that such counts are, by default, incompletely reported: a number without its sampling budget carries less information than authors and readers typically treat it as carrying.

2 Related work

Comparative benchmarks of dimensionality reduction for molecular systems. Recent work has directly compared PCA, TICA, and VAE projections [1, 2, 3, 4, 5, 6], together with downstream clustering choices, on protein folding trajectories and on Markov state model construction [7, 8, 9], finding that the choice of method changes the resulting free energy surface and the states it appears to reveal; Glielmo et al. [10] survey the field. This literature establishes that method choice matters at fixed data. The present work holds method fixed and varies the data, which is the complementary axis: we ask not which method is best at a given budget, but whether the answer any method gives depends on the budget itself. We are not aware of prior work that varies sampling budget at fixed, independently known ground truth to isolate this effect.

Current best-practice tutorials explicitly separate dimensionality reduction and geometric clustering from kinetic validation, including lag sensitivity, implied-timescale analysis and Chapman-Kolmogorov testing [11]. That distinction sets the scope of this paper: its fitted GMM components are candidate structural states, not a substitute for a validated kinetic model.

Undersampling and non-ergodicity in molecular dynamics. It is well established and frequently acknowledged that molecular dynamics trajectories are undersampled relative to the timescales of interest, and that this undersampling can bias downstream analysis [12]. This is typically stated as a caveat rather than measured as a quantitative confound with a known ground-truth comparison. Our contribution is to make the undersampling question tractable by constructing systems where the true answer is known, and to show that the resulting effect is not explained by exploration deficit alone, since it persists when landscape coverage is matched across budgets.

Model selection under misspecification. The behavior of BIC and related criteria when the fitted model class does not match the true data-generating process is well studied in the statistics literature [13, 14, 15]; in particular, the likelihood gain from additional mixture components can grow with sample size under misspecification even when the penalty term

does not compensate proportionally. Our results are consistent with this mechanism contributing to the effect, but not with it being the whole explanation: the effect’s persistence under coverage matching, and its variation across a projection method (TICA) whose own stability is separately budget-dependent, point to more than one contributing cause. We are explicit throughout about which part of the effect is attributable to which mechanism, rather than presenting a single unified explanation the data does not fully support.

We are generous to this prior work by design. The gap we identify is narrow: existing benchmarks are good benchmarks, conducted at whatever budget the authors used, and simply do not vary that budget as an independent condition against known ground truth. The contribution of this paper is the control, not a correction to any specific published result.

3 Setup

3.1 Systems

Three systems are used. The synthetic systems have exact basin labels; alanine uses an independently specified dihedral reference partition rather than a kinetically validated ground truth.

Müller-Brown potential (2 latent dimensions, 3 basins). A standard three-well potential [16] with minima located by gradient descent from known starting points; basin membership is assigned by steepest-descent quenching to the nearest minimum, never by a fitted clustering model. Figure 1 shows the potential surface, the resulting ground-truth basins, and a representative sampled trajectory.

Prinz potential (1 latent dimension, 4 basins). A standard four-well potential from the Markov state model literature [17]; basin boundaries are the potential’s barrier tops, located once on a fine grid at construction time.

Alanine dipeptide (all-atom, 3 reference regions). Two independent 100 ns Langevin dynamics simulations in implicit solvent (OpenMM [18], amber14 [19] with GBn2 [20], 2 fs timestep, 1 ps save interval, 100,000 saved frames each). The reference labels use a fixed three-region partition of backbone (ϕ , ψ) space—a chemical definition entirely independent of dimensionality reduction or clustering. Observed features are pairwise distances between heavy atoms, deliberately excluding ϕ/ψ , so that the methods under audit must recover the landscape from a representation that does not already contain the answer. The simulation length was chosen after an initial 5 ns pilot run showed the rarest basin ($\alpha_L/C7ax$) was visited inconsistently at small sampling budgets; 100 ns ensures all three basins are visited at every tested budget under coverage-matched subsampling, and at budgets of 8,000 frames and above under short trajectories (Section 7).

For the two synthetic potentials, the low-dimensional dynamics are lifted into 30 observed dimensions through a fixed random Fourier feature map plus isotropic noise, drawn once and held constant across every sampling condition, so that no difference between conditions is attributable to a changing observation model. Figure 2 shows the resulting embeddings

Ground truth is known by construction, not inferred

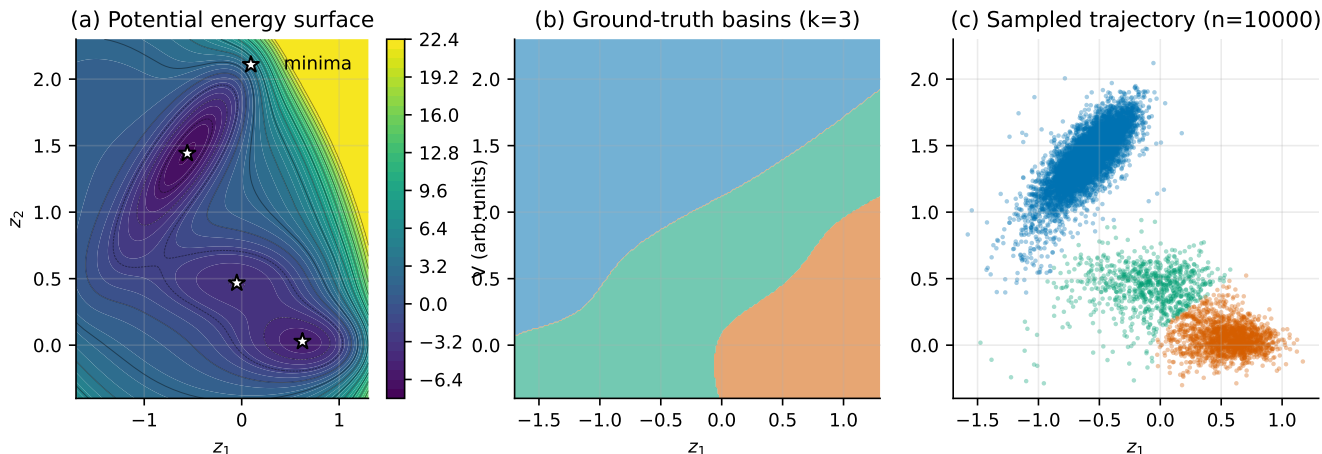


Figure 1: Ground truth for the Müller–Brown potential: (a) the energy surface with its three minima, (b) basin assignment by steepest-descent quenching, (c) one sampled trajectory coloured by true basin. Basin labels never come from a fitted model.

across projection method and sampling budget on Müller–Brown, coloured by true basin, to illustrate the visual character of the confound before the quantitative results in Sections 4-6: the underlying landscape is identical in every panel.

3.2 Sampling conditions

Two modes are compared at each of several sampling budgets (250 to 32,000 frames for the synthetic potentials; 100 to 64,000 frames for alanine dipeptide, whose 100,000-frame trajectory supports a wider sweep than either synthetic system):

- **short:** a trajectory run to produce exactly the target number of saved frames. This is the condition an analyst with a fixed compute budget actually has, and it confounds two things: how much of the landscape was explored, and how many points the selection criterion sees.
- **subsample:** one long reference trajectory per seed, uniformly thinned to the target budget. Landscape coverage is held approximately fixed by construction across every budget; only the number of points varies.

Comparing the two isolates whether an observed effect is due to incomplete exploration (would appear in *short* but not *subsample*) or to the estimator’s response to sample size (would appear in both).

An additional IID equilibrium control removes both time correlation and projection learning. For each synthetic potential, independent samples are drawn with replacement from a fine-grid Boltzmann distribution in the exact latent coordinates (20,001 grid points in one dimension; 240 x 240 in two), using 10 seeds and the same budgets and BIC search ceiling. This control asks whether BIC component splitting requires either a trajectory or a learned embedding.

3.3 Methods under audit

PCA [1], **TICA** [2, 3] (solved as an explicit generalized eigenproblem so its regularization is auditable rather than a library default), and a **VAE** [21, 4] (fixed architecture, trained to a fixed gradient-step target rather than a fixed epoch count). Full epochs cause 0-32 extra updates depending on budget, at most 1.1% of the 3,000-4,000-step target; this disclosed rounding variation is not proportional to dataset size. t-SNE [22] is excluded from the quantitative audit because its apparent cluster structure depends strongly on perplexity and other visualization choices [23].

3.4 State-count selection

Five diagnostics for choosing the number of components in the projected space are compared on identical embeddings: BIC [13], AIC [24], the integrated completed likelihood (ICL [25], which adds an entropy penalty on component overlap to BIC), silhouette score [26], and an elbow heuristic on k-means inertia. BIC, AIC and ICL select Gaussian-mixture models. Silhouette and the simple largest-relative-inertia-drop elbow rule are descriptive sensitivity checks, not validated state estimators; their selected k is used to refit a GMM only so recovery is measured on a common clustering family. The comparison is used to distinguish criterion-specific failure modes, not to claim that five independent estimators corroborate one mechanism.

3.5 Metrics and statistics

Recovery is measured by the adjusted Rand index (ARI) [27] between the recovered clustering and ground-truth basin membership, both at the criterion-selected state count and, as a control, at the true state count supplied directly (“oracle k ”). ARI is chance-corrected, so it does not reward a method for

Embeddings coloured by TRUE basin; the landscape is identical in every panel

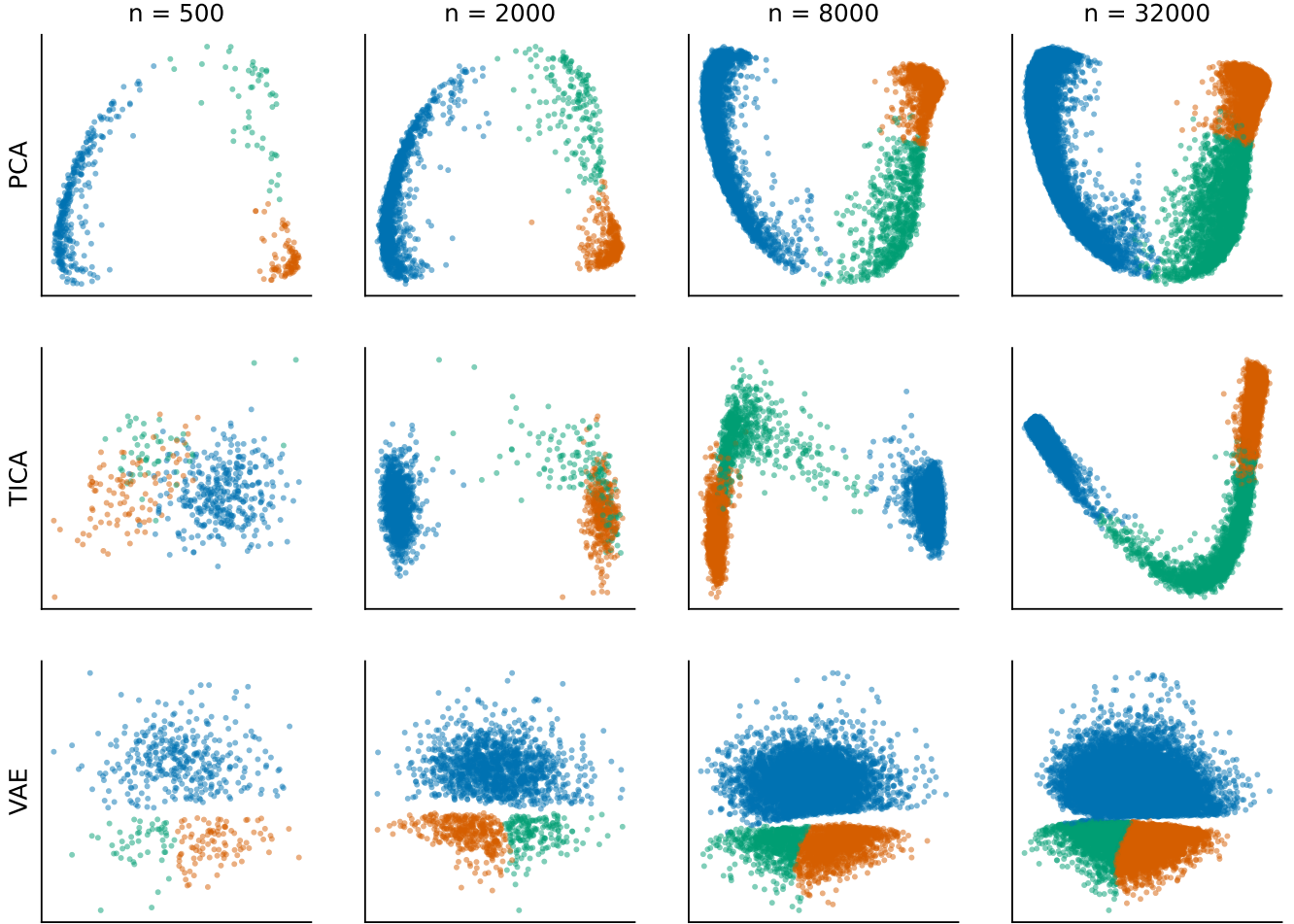


Figure 2: Embeddings of the Müller–Brown system across projection method (rows) and sampling budget (columns), coloured by true basin. The underlying landscape is identical in every panel; only the number of saved frames differs.

reporting more clusters—the specific failure mode under investigation. **Basin visitation** and **rarest-basin occupancy** are tracked per condition to identify and exclude degenerate cases in which a trajectory visited fewer true basins than exist, which produce a constant ground-truth label vector against which any single-cluster solution scores $\text{ARI} = 1.0$ trivially.

The seed is the unit of replication: frames within a trajectory are strongly autocorrelated, so all confidence intervals are bootstrapped over seeds, never over frames. The headline statistic reported for the budget effect is a Spearman rank correlation between log sampling budget and reported state count, chosen because the claim under test is monotone inflation rather than any specific functional form. Twenty seeds are used per condition on the synthetic potentials; eight seeds on alanine dipeptide, reflecting its higher per-condition compute cost.

For the coverage contrast, seeds are paired across *short* and *subsample* and resampled jointly. The reported inter-

val is for $\rho(\text{subsample})$ minus $\rho(\text{short})$; overlap between two marginal confidence intervals is not used as a hypothesis test. Pooled row-level p-values are not interpreted because repeated budgets within a seed are not independent replicates.

4 Reported state count tracks sampling budget

Across both landscapes, all three projections, and both sampling modes, the number of Gaussian components selected by BIC increases strongly with the number of saved frames, while the underlying potential is identical in every condition.

On Müller–Brown (true $k = 3$), BIC-selected state counts rise from 1.0–6.3 at $n = 250$ to 11.9–14.8 at $n = 32,000$. On Prinz (true $k = 4$) the same sweep runs from 1.0–7.1 up to 12.5–14.9. At the largest budget every method on every system reports at least three times the true number of basins.

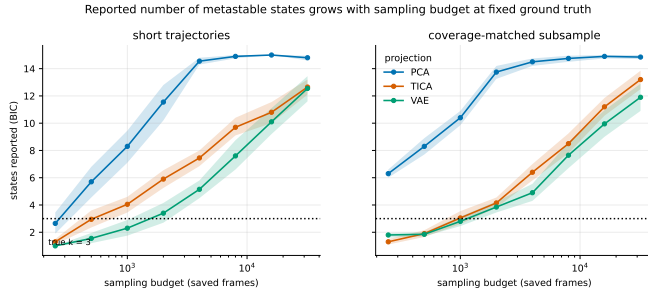


Figure 3: BIC-selected state count against sampling budget, with bootstrap confidence intervals over seeds. The dotted line is the true state count. Counts rise monotonically while the landscape is unchanged.

The rank correlation between log sampling budget and BIC-selected state count is uniformly high and tightly bounded (Table 1):

Bootstrap intervals are over the 20 seeds. No interval approaches zero. (Figure 3)

Two points of interpretation are worth making immediately, because they bound what this result does and does not say.

First, the effect is **not** a claim that these projections fail to represent the landscape. Section 6 shows the opposite. It is a claim about the composite pipeline of projection followed by model selection, which is how state counts are obtained in practice.

Second, the reported counts at large budgets are **right-censored**. The search range was capped at $k = 15$. Under BIC, PCA hits that cap in 44% of Müller-Brown conditions and 29–30% of Prinz conditions (Table 2). Where the cap binds, the tabulated mean is a lower bound on what the criterion would have selected, and the rank correlation is correspondingly attenuated. The reported effect is therefore conservative.

5 Selection criteria exhibit different failure modes

The obvious objection is that BIC’s penalty grows as $\log n$ while the likelihood gain from an additional mixture component grows with n , so BIC selects more components as n increases under any model misspecification — and a projected free energy basin is never exactly Gaussian. On that account the result would be a property of the criterion rather than of the analysis pipeline.

Five criteria were therefore computed on every condition. The outcome is more informative than a simple yes or no.

AIC is frequently ceiling-censored and non-monotone. AIC reaches the search ceiling in 24–84% of Müller-Brown conditions and 12–65% of Prinz conditions (Table 2) — 65–84% for PCA specifically on Müller-Brown, 24–28% for TICA and VAE there—but Figure 4 shows why that number cannot be read as inflation: AIC’s curve is U-shaped, starting at 14 states for PCA and 8 for TICA and VAE at $n = 250$, dipping near $n = 1000$, then returning to 14.5. It over-selects at every budget,

including the smallest. Its rank correlation is computed across a non-monotone curve that spends most of its range censored at the cap and is therefore uninterpretable. AIC is reported for completeness and excluded from the trend claim.

ICL does not reach the tested ceiling, but it is not accurate. ICL is the only criterion that reached the ceiling in zero of twelve (system $\times$ mode $\times$ method) cells, which keeps its output in an interpretable range. Within that range it is wrong in a method-dependent direction (Figure 4, Müller-Brown, true $k = 3$): PCA reports 5.0–6.2 states throughout, roughly twice the truth and nearly flat; VAE reports 1.2–1.8, *under-selecting* at every budget; only TICA rises monotonically through the true value, from 1.0 to 4.6. The defensible claim is that ICL avoids the runaway behaviour of BIC and AIC, not that it recovers the correct state count.

Silhouette and elbow gap are insensitive rather than robust. Both return a near-constant answer regardless of budget. PCA’s silhouette selection is exactly 2 at every budget on Müller-Brown and exactly 4 at every budget on Prinz; elbow gap is flat at 2 for PCA and VAE on both systems. The Prinz case is the more instructive one: silhouette’s constant output there coincides with the true state count, so on that system alone the criterion appears to perform perfectly. It has not measured anything. A criterion that returns the same answer irrespective of the data will look excellent on any landscape where that answer happens to be correct, and this is precisely how an uninformative criterion can acquire a reputation for reliability. The sign changes in Table 3 (silhouette/PCA: -0.26 on Müller-Brown, +0.65 on Prinz) are sign changes on an essentially constant output and should be read as noise, not as landscape sensitivity.

The honest summary is narrower than “five criteria disagree” and narrower than a clean ordering by penalty strength: **BIC inflates monotonically with sampling budget; AIC is saturated at all budgets; ICL alone stays in an interpretable range without saturating, though it remains inaccurate in a method-dependent direction; and the two geometry-based criteria are unresponsive to budget and to landscape alike.** (Figure 4)

5.1 The two supporting tables

Ceiling-hit rate—the fraction of conditions in which the criterion selected the largest state count the search allowed ($k_{\max} = 15$). Where this is non-negligible the reported count is right-censored and the inflation is understated (Table 2):

Budget correlation under each criterion, as Spearman ρ between log sampling budget and selected state count. Entries marked “–” are conditions in which the criterion returned the same state count at every budget, so the correlation is undefined—itself the clearest statement of how uninformative silhouette and elbow gap are here:

5.2 One exception, stated plainly

ICL’s resistance does not extend to TICA. TICA’s ICL correlation is +0.74 and +0.92 on Müller-Brown (short, subsample)

Table 1: Spearman rank correlation between log sampling budget and BIC-selected state count, for each system, sampling mode and projection method. Brackets are bootstrap 95% confidence intervals over the 20 seeds. No interval approaches zero.

system	mode	PCA	TICA	VAE
Müller-Brown	short	+0.88 [+0.85, +0.91]	+0.94 [+0.93, +0.96]	+0.92 [+0.91, +0.94]
Müller-Brown	subsample	+0.88 [+0.85, +0.91]	+0.96 [+0.95, +0.97]	+0.93 [+0.92, +0.94]
Prinz	short	+0.95 [+0.93, +0.96]	+0.97 [+0.97, +0.98]	+0.93 [+0.92, +0.94]
Prinz	subsample	+0.95 [+0.94, +0.96]	+0.96 [+0.95, +0.97]	+0.91 [+0.89, +0.93]

Table 2: Ceiling-hit rate: the fraction of conditions in which the criterion selected the largest state count the search allowed ($k_{\max} = 15$). Where this is non-negligible the reported count is right-censored, so the inflation reported elsewhere is understated. ICL is the only criterion that never reaches the ceiling.

system	mode	criterion	PCA	TICA	VAE
Müller-Brown	short	BIC	44%	1%	4%
Müller-Brown	subsample	BIC	42%	3%	2%
Müller-Brown	short	AIC	66%	27%	24%
Müller-Brown	subsample	AIC	84%	25%	28%
Müller-Brown	short	ICL	0%	0%	0%
Müller-Brown	subsample	ICL	0%	0%	0%
Müller-Brown	short	silhouette	0%	1%	2%
Müller-Brown	subsample	silhouette	0%	1%	0%
Müller-Brown	short	elbow gap	0%	0%	0%
Müller-Brown	subsample	elbow gap	0%	0%	0%
Prinz	short	BIC	29%	0%	9%
Prinz	subsample	BIC	30%	0%	8%
Prinz	short	AIC	62%	19%	37%
Prinz	subsample	AIC	65%	12%	36%
Prinz	short	ICL	0%	0%	0%
Prinz	subsample	ICL	0%	0%	0%
Prinz	short	silhouette	0%	0%	0%
Prinz	subsample	silhouette	0%	1%	0%
Prinz	short	elbow gap	0%	0%	0%
Prinz	subsample	elbow gap	0%	0%	0%

and +0.91 and +0.93 on Prinz — comparable to its BIC correlations rather than to PCA’s and VAE’s ICL values.

We do not have a confirmed mechanism for this. One plausible account is that TICA components, being constructed to maximise autocorrelation rather than variance, produce latent densities less well approximated by a Gaussian mixture, weakening the entropy penalty that ICL relies on. This is a conjecture and is flagged as such; it is a natural target for follow-up rather than something the present data settles. It is reported here because a recommendation to use ICL would be misleading without it.

6 Incomplete exploration is not a sufficient explanation

The competing mechanism is exploration deficit: short trajectories visit less of the landscape, so fewer or differently shaped basins are populated, and the apparent structure changes for reasons that have nothing to do with sample size.

The `subsample` mode addresses this directly. A single long reference trajectory per seed is thinned uniformly to each

target budget, so landscape coverage is held approximately fixed by construction while only the number of points varies.

Coverage matching does not remove the effect. BIC budget correlations under `short` and `subsample` are closely similar in magnitude (Table 4):

Paired seed bootstraps estimate `subsample` minus `short` ρ . All absolute differences are at most 0.022; two TICA intervals exclude zero, so the modes are not literally statistically indistinguishable. The practically small deltas show that strong positive BIC trends survive approximate coverage matching, which is inconsistent with incomplete exploration as the sole explanation. (Figure 5)

The effect also survives independent equilibrium sampling in the exact latent coordinates. This control removes temporal correlation, incomplete trajectory mixing, observation noise, and dimensionality reduction simultaneously. BIC still splits the fixed equilibrium densities increasingly finely as sample size grows. The independent-sample result is (Table 5):

Intervals bootstrap the 10 independent IID samples. Because the inputs are the exact latent coordinates and contain no time ordering, this result identifies BIC under a misspeci-

Table 3: Budget correlation under each selection criterion: Spearman ρ between log sampling budget and selected state count. Entries marked “–” are conditions in which the criterion returned the same count at every budget, leaving the correlation undefined.

system	mode	criterion	PCA	TICA	VAE
Müller-Brown	short	BIC	+0.88	+0.94	+0.92
Müller-Brown	subsample	BIC	+0.88	+0.96	+0.93
Müller-Brown	short	AIC	+0.61	+0.71	+0.75
Müller-Brown	subsample	AIC	+0.31	+0.67	+0.64
Müller-Brown	short	ICL	+0.58	+0.74	+0.41
Müller-Brown	subsample	ICL	+0.34	+0.92	+0.28
Müller-Brown	short	silhouette	-0.26	+0.14	-0.52
Müller-Brown	subsample	silhouette	–	+0.30	+0.03
Müller-Brown	short	elbow gap	+0.03	+0.37	+0.46
Müller-Brown	subsample	elbow gap	+0.00	+0.54	-0.03
Prinz	short	BIC	+0.95	+0.97	+0.93
Prinz	subsample	BIC	+0.95	+0.96	+0.91
Prinz	short	AIC	+0.67	+0.82	+0.77
Prinz	subsample	AIC	+0.66	+0.85	+0.74
Prinz	short	ICL	+0.48	+0.91	+0.46
Prinz	subsample	ICL	+0.36	+0.93	+0.24
Prinz	short	silhouette	+0.65	+0.26	+0.41
Prinz	subsample	silhouette	+0.12	-0.22	+0.12
Prinz	short	elbow gap	-0.48	+0.28	-0.05
Prinz	subsample	elbow gap	-0.12	+0.09	-0.13

Table 4: The coverage-matched control. BIC budget correlations under genuinely short trajectories and under uniform thinning of one long reference trajectory. The paired column is subsample minus short ρ , bootstrapped over matched seeds. All absolute differences are at most 0.022, although two TICA intervals exclude zero.

system	method	short	subsample	paired delta ρ [95% CI]
Müller-Brown	PCA	+0.88 [+0.85, +0.91]	+0.88 [+0.85, +0.91]	+0.002 [-0.037, +0.037]
Müller-Brown	TICA	+0.94 [+0.93, +0.96]	+0.96 [+0.95, +0.97]	+0.017 [+0.004, +0.030]
Müller-Brown	VAE	+0.92 [+0.91, +0.94]	+0.93 [+0.92, +0.94]	+0.007 [-0.014, +0.028]
Prinz	PCA	+0.95 [+0.93, +0.96]	+0.95 [+0.94, +0.96]	+0.004 [-0.011, +0.020]
Prinz	TICA	+0.97 [+0.97, +0.98]	+0.96 [+0.95, +0.97]	-0.014 [-0.024, -0.005]
Prinz	VAE	+0.93 [+0.92, +0.94]	+0.91 [+0.89, +0.93]	-0.022 [-0.047, +0.001]

fied finite-Gaussian model as sufficient to produce a substantial budget trend. It does not show that temporal dependence and projection quality are irrelevant in the trajectory pipelines; those factors can modify the magnitude and, for TICA, the low-budget behavior.

6.1 Recovery degrades as sampling budget increases

The gap between recovery at the selected state count and recovery at the true state count is not constant. Recovery under the standard pipeline peaks at an intermediate budget and then declines, while oracle-k recovery over the same range is flat or rising. (Figure 6)

A coverage artifact was checked for before drawing this conclusion. Conditions with incomplete reference-basin support can make recovery misleading; the one-basin special case produces a constant reference-label vector against which a one-cluster solution scores $\text{ARI} = 1.0$ trivially. Such conditions occur only in `short` mode at small budgets—100% of

Müller-Brown conditions at $n = 250$, declining to zero by $n = 4000$, and 35% of Prinz conditions at $n = 250$, zero by $n = 1000$ —and in `no subsample` condition on either system. All 38 rows scoring $\text{ARI} > 0.999$ are degenerate. Because coverage-matched subsampling contains no degenerate conditions, the primary curves are unaffected: cleaned and raw values agree to two decimal places throughout. Excluding degenerate rows from `short` mode makes the decline steeper rather than shallower (Müller-Brown TICA: peak $0.61 \rightarrow 0.76$, final drop -0.42 ; VAE final drop -0.61), because those rows carried low median ARI (0.023) at the small-budget end. The Müller-Brown `short` condition at $n = 250$ is excluded entirely, having no valid replicates.

Every method-system pair on the two synthetic potentials shows the peak-then-decline pattern under coverage matching; the complete picture across all three systems, including alanine dipeptide, is given as a single table in Section 6.3 rather than duplicated here.

For these pairs, collecting more data makes the answer produced by the standard pipeline substantially worse, with the

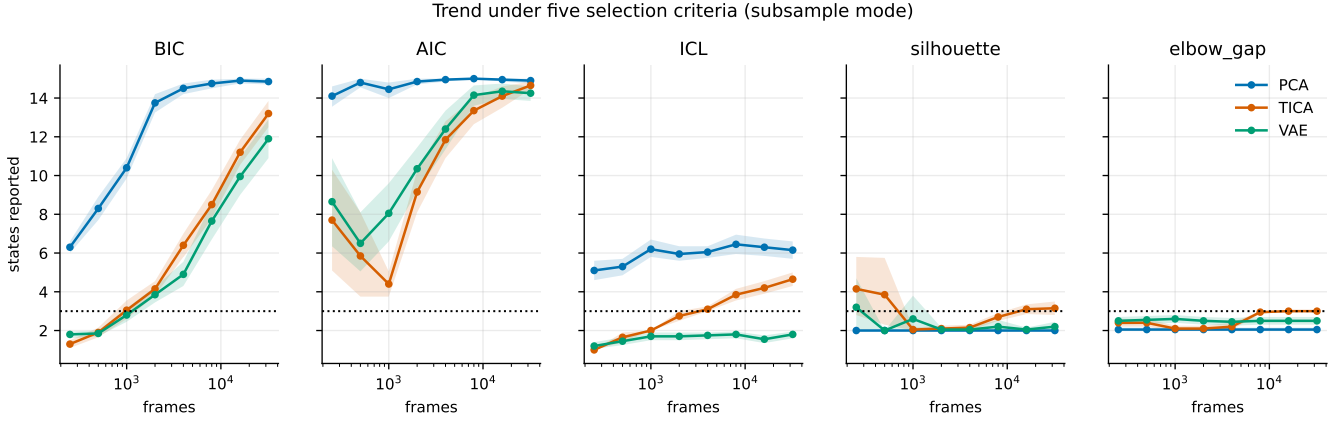


Figure 4: Criterion-dependent behavior on identical embeddings. BIC inflates monotonically; AIC is frequently ceiling-censored and non-monotone; ICL stays within the tested range; silhouette and the simple elbow diagnostic are largely insensitive.

Table 5: Independent-equilibrium control in the exact latent coordinates (10 seeds). Samples are drawn independently from a gridded Boltzmann distribution, removing trajectory autocorrelation and dimensionality reduction. BIC component counts still rise with budget.

system	BIC budget ρ [95% CI]	mean k at n=250	mean k at n=32,000
Müller-Brown	+0.92 [+0.90, +0.94]	2.7	8.1
Prinz	+0.86 [+0.84, +0.88]	4.0	7.0

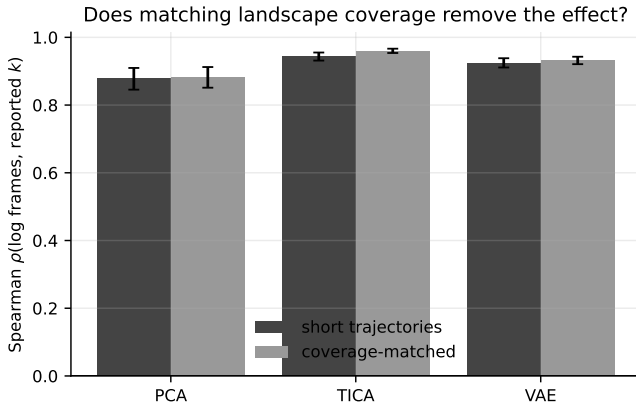


Figure 5: Budget correlation with and without coverage matching. The effect survives holding landscape exploration fixed, which places it in the estimator rather than in what the trajectory visited.

crossover falling between $n = 250$ and $n = 1000$. This is stronger than the endpoint comparison in Table 7, which reports only the largest budget and therefore conceals the non-monotonicity.

6.2 The TICA exception, and the evidence for its mechanism

TICA on Prinz is the exception: recovery rises from 0.00 at $n = 250$ to 0.58 at $n = 8000$ before a small decline to 0.51. It is not an instance of the pattern and is not presented as one.

The mechanism is directly tested, not only inferred from this exception. A lag/lagged-pair ablation (Prinz potential, lags in

{5, 10, 20, 50, 100}, 10 seeds, subsample mode, 300 conditions) fits TICA at every combination of sampling budget and lag and measures oracle-k recovery—the projection’s quality with the selection step removed—as a function of both (Table 6):

Recovery generally decreases as lag increases at fixed budget, with small non-monotone reversals at low recovery. Lag changes the dynamical covariance target as well as reducing the number of usable pairs from n_{frames} to $n_{\text{frames}} - \text{lag}$; because that pair-count reduction is modest at the largest budgets, pair count alone cannot explain the effect. A 0.35 oracle-k ARI is reached by $n = 1000$ at lag = 5, around $n = 8000$ at lag = 50, and only at $n = 16000$ at lag = 100. The ablation establishes lag sensitivity and a lag-dependent data requirement, but not a single causal mechanism. The observed curve is consistent with two opposing budget effects—lag-dependent projection recovery and a selection step that degrades with data after recovery becomes high (visible in the corresponding selected-k table, where recovery at lag = 5 peaks at $n = 8000$, 0.56, and declines to 0.45 by $n = 16000$, reproducing the main peak-then-decline pattern within a single fixed lag setting). Where the crossover falls depends on how quickly TICA stabilises on that landscape’s combination of lag and dynamics: near $n = 1000$ on Müller-Brown at the default lag, near $n = 8000$ on Prinz. PCA and VAE have no comparable warm-up requirement, consistent with their declining monotonically or peaking at the smallest budgets in the main sweep.

6.3 Scope of the localisation claim

Oracle-k recovery separates cleanly by system rather than by method. On Prinz, all three projections reach 0.81-0.90 when

Table 6: TICA lag ablation on the Prinz potential: oracle- k recovery as a function of sampling budget and lag (10 seeds, subsample mode, 300 conditions). Recovery generally falls as lag grows, with small low-recovery reversals; the budget needed to reach a given level rises.

n_frames	lag=5	lag=10	lag=20	lag=50	lag=100
500	0.10	0.03	0.03	0.05	0.03
1000	0.35	0.16	0.06	0.06	0.07
2000	0.53	0.40	0.24	0.05	0.06
4000	0.67	0.55	0.42	0.20	0.09
8000	0.77	0.69	0.59	0.44	0.29
16000	0.87	0.80	0.70	0.57	0.54

given the true state count, so the projections are faithful and essentially all of the observed degradation is attributable to state-count selection. On Müller-Brown only TICA reaches that range (0.85); PCA and VAE reach only 0.41-0.44 even with the true k supplied, so for those pairs the projection is itself a limiting factor and selection is not the whole story.

The claim that the projection is faithful and only the selection step fails is therefore supported wherever oracle- k recovery is high—all of Prinz, and TICA on Müller-Brown—and is not supported in general. The paper should scope it accordingly rather than asserting it as a universal result.

(All values at $n = 32,000$, `short` mode; `subsample` values are within 0.03. Note that these endpoint figures understate oracle- k recovery for TICA under subsampling, which reaches 0.82 on Prinz at the largest budget.)

6.4 The complete picture across nine method-system pairs

Every method-system pair tested shows recovery declining from its peak to the largest budget tested. What differs is how much, and the amount tracks a specific mechanism rather than varying arbitrarily.

Two patterns are visible in this table together that were not visible when the systems were reported separately.

PCA and VAE decline sharply and consistently (43-84% relative decline) across all three systems, regardless of how faithful the underlying projection is (compare their oracle- k column, which ranges from 0.42 to 0.90). Their selected- k recovery is dominated by the state-count selection failure, not by projection quality.

TICA’s decline is associated with how much budget remains after its high-recovery crossover. TICA requires enough data at the chosen lag to produce a stable projection at all; at small budgets on every system it starts near zero (Müller-Brown: 0.09 at $n=250$; Prinz: 0.00 at $n=250$ -500; alanine: 0.00 at $n=100$). The diagnostic used here for entry into high recovery is the oracle- k column, which measures projection quality with the selection step removed. We define entry into the high-recovery regime *a priori* as mean oracle- k ARI at least 0.80. With that transparent threshold, the subsequent decline is associated with how many budget doublings remain (Table 9):

Prinz spends its entire range approaching high recovery and barely declines; Müller-Brown crosses early and declines as

sharply as PCA and VAE on the same system; alanine dipeptide falls between them because its threshold occurs two doublings before the sweep does. Across only three systems this is an association, not a validated law. It is consistent with two mechanisms operating on one curve rather than one: an estimator-stability effect that dominates at small budgets, and a selection-driven degradation that dominates once the estimator has stabilized, exactly as characterized via the oracle- k contrast in Section 6.

Why alanine enters the high-recovery regime late, and a caution it implies. That alanine crosses the 0.80 threshold at $n=16,000$ is at first surprising, since its reference trajectory is the longest in the paper. The association with effective lag is specific to the `subsample` condition. TICA’s lag is specified in frames of the data it is handed, but uniform thinning to a budget of n frames from a 100,000-frame trajectory imposes a stride of $100,000/n$, so a nominal lag of 10 frames corresponds to an *effective* physical lag of $10 \times 100,000/n$ picoseconds. The effective lag therefore shrinks as the budget grows, and TICA’s projection quality tracks it closely (Table 10):

Oracle- k recovery is flat at roughly 0.25 across six consecutive budgets and then rises nearly fourfold between $n=8,000$ and $n=16,000$ —near where the effective lag crosses below about 100 ps. This is an empirical crossover in the present sweep, not an independently fitted dihedral transition timescale. Changing effective lag changes the time-lagged covariance target; the present data do not identify one unique causal explanation. The same figures in `short` mode, where the stride is always 1 and the effective lag is fixed at 10 ps, show no such discontinuity: oracle- k rises smoothly from 0.15 to 0.96. Figure 7 plots both modes against budget and against effective lag; the discontinuity is present in the first view and resolved by the second.

The practical caution follows directly, and it is not one we set out to test: thinning a trajectory before applying TICA rescales the lag along with the data. An analyst who subsamples aggressively and keeps the lag parameter fixed is silently lengthening the effective lag, and can push a well-sampled trajectory back into a low-recovery regime. This is a complementary lag-sensitivity observation on real molecular dynamics, arrived at through the thinning stride rather than through a direct parameter sweep.

All nine method-system pairs in the seed-0 summary decline from peak to the largest budget tested. For TICA, decline magnitude is associated with a specific, testable crossover ac-

Table 7: Recovery at the criterion-selected state count against recovery given the true state count, at $n = 32,000$ in `short` mode. The gap localises how much of the error belongs to the selection step rather than to the projection.

system	method	selected k	oracle k	gap
Müller-Brown	PCA	0.102	0.415	0.31
Müller-Brown	TICA	0.342	0.847	0.51
Müller-Brown	VAE	0.111	0.423	0.31
Prinz	PCA	0.349	0.897	0.55
Prinz	TICA	0.482	0.815	0.33
Prinz	VAE	0.399	0.854	0.46

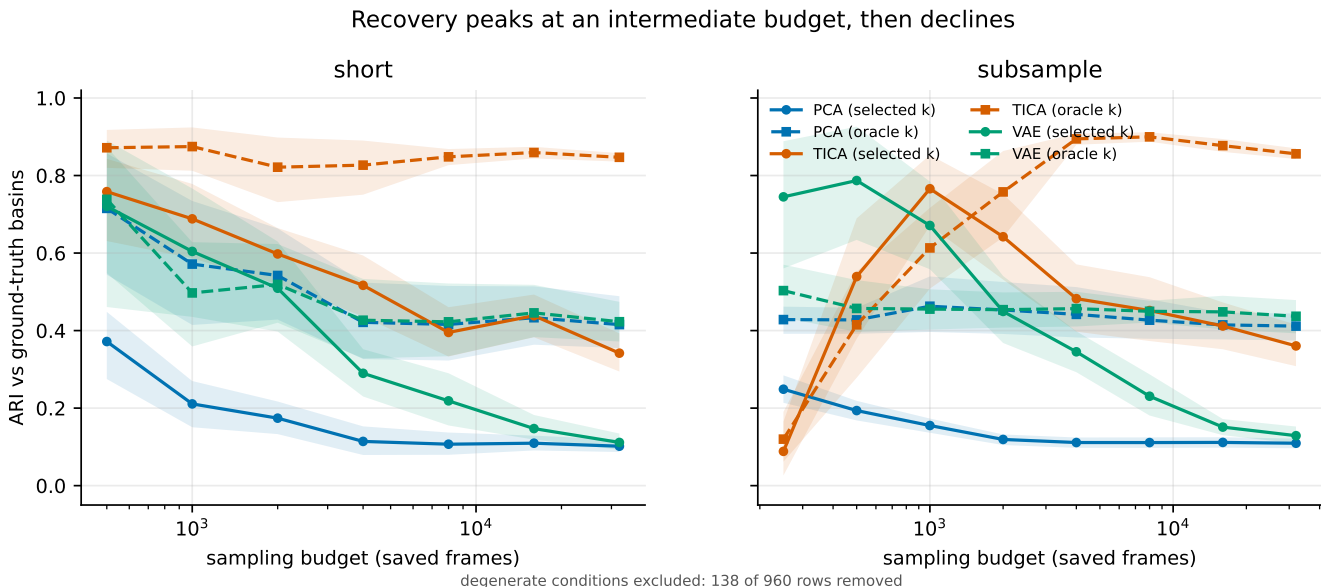


Figure 6: Recovery at the selected state count against recovery given the true state count. Selected- k recovery peaks at an intermediate budget and then declines while oracle- k recovery is flat or rising: more data makes the standard pipeline’s answer worse. Degenerate conditions, in which fewer basins were visited than exist, are excluded.

count, subject to the cross-seed qualification below. This table, not a summary fraction, is the result that belongs in the paper.

One qualification, established by the second alanine seed reported in Section 7 and not visible in this table. The decline column for TICA depends on where that system enters the high-recovery regime, and the crossover budget is itself seed-dependent: on a second independent 100 ns trajectory TICA crosses one doubling later, at $n=32,000$, and consequently shows no decline within the tested range. This is consistent with zero doublings remaining, as on Prinz, but it means the TICA rows here should be read as one draw from a seed-dependent quantity rather than as a fixed property of the system. The PCA and VAE rows replicate closely across seeds.

7 Alanine dipeptide: simulation detail and comparison to the synthetic systems

The synthetic systems establish the effect under conditions where ground truth is exact and the observation model is fully controlled. This section reports the real-molecule replication and what is and is not comparable between the two settings.

Simulation. Two independent 100 ns trajectories of alanine dipeptide (ACE-ALA-NME) was run in implicit solvent (OpenMM [18], amber14 [19] with the GBn2 implicit solvent model [20], Langevin middle integrator [28], 2 fs timestep, 300 K, 1 ps save interval), producing 100,000 saved frames. A second trajectory was run identically from a different random seed; unless stated otherwise the results below are from the first, and the two are compared at the end of this section. Dihedrals and heavy-atom distances were computed with MDTraj [29]. An initial 5 ns pilot run was discarded after it showed the rarest of the three basins ($\alpha_L/C7ax$) was not consistently visited at the sampling budgets under study; 100 ns was chosen because it visits all three basins at every tested budget

Table 8: Seed-0 summary of all nine method–system pairs. Every pair declines from its peak to the largest budget tested. PCA and VAE declines are uniformly large; the TICA association is qualified by the independent seed comparison.

system	method	peak ARI (budget)	ARI at largest budget	relative decline	oracle-k ARI
Müller-Brown	PCA	0.25 (n=250)	0.11	56%	0.415
Müller-Brown	TICA	0.77 (n=1000)	0.36	53%	0.847
Müller-Brown	VAE	0.79 (n=500)	0.13	84%	0.423
Prinz	PCA	0.64 (n=250)	0.35	45%	0.897
Prinz	TICA	0.58 (n=8000)	0.51	12%	0.815
Prinz	VAE	0.72 (n=1000)	0.41	43%	0.854
Alanine dipeptide	PCA	0.47 (n=100)	0.12	74%	0.475
Alanine dipeptide	TICA	0.44 (n=16000)	0.27	37%	0.963
Alanine dipeptide	VAE	0.54 (n=2000)	0.18	66%	0.421

Table 9: Descriptive TICA crossover comparison. Entry into a high-recovery regime is defined as mean oracle- k ARI at least 0.80. Across three systems, later entry is associated with less subsequent decline; this is not a validated law.

system	first oracle-k ARI ≥ 0.80	doublings remaining in sweep	decline
Prinz	n=32,000 (end of range)	0	12%
Alanine dipeptide	n=16,000	2	37%
Müller-Brown	n=1,000	5	53%

under coverage-matched subsampling, and at budgets of 8,000 frames and above under short trajectories (Table 11). Sampling budgets from 100 to 64,000 frames were drawn from each trajectory: `short` budgets as random contiguous windows, `subsample` budgets as uniform thinning, matching the two conditions used on the synthetic systems. Reference-region membership was assigned from backbone (ϕ , ψ) dihedrals using a fixed three-region partition (C7eq, α_R , α_L /C7ax), independent of any method under audit. These labels are a structural reference, not proof that the regions are kinetically metastable. Figure 8 shows, for the seed-1 trajectory, that partition and the resulting basin populations: occupancy is strongly uneven, at 0.639, 0.308 and 0.053, and it is the 5.3% basin that makes the sampling budget bite. These exact occupancies are seed-1-specific and are not asserted to represent seed 0; the cross-seed statement is limited to whether all three regions are visited. At $n = 100$ frames a basin that rare is expected to contribute about five frames, which is why the 5 ns pilot reached it inconsistently and why coverage matching matters so much on this system. Observed features for the dimensionality reduction methods were pairwise distances between heavy atoms, excluding ϕ/ψ so that the audited methods must recover the landscape from a representation that does not already encode the answer.

Under coverage-matched subsampling, all three basins are visited at every budget tested, from $n=100$ upward, since subsampling draws uniformly from the full 100,000-frame reference trajectory rather than from a contiguous block.

Replication of the main effect. BIC-selected component count increases with sampling budget on all three methods, in both modes (Spearman $\rho = 0.87$ - 0.97 , 8 seeds per correlation—fewer than the 20 seeds used on the synthetic systems, per the limitation discussed in Section 9). PCA rises from a mean of 3.1 states at $n=100$ to 14.5 at $n=64,000$, against

a true count of 3. TICA and VAE rise from conservative counts near 1 at the smallest budget to 13.9 at the largest. All three methods approach the search ceiling ($k_{\max}=15$) at the largest budgets tested, consistent with the ceiling-censoring pattern documented on the synthetic systems.

A method ranking that reverses relative to the synthetic systems. On the synthetic potentials, TICA is not uniformly the best-recovering method (it matches PCA and VAE’s oracle- k quality only on Müller-Brown and is comparable to all three on Prinz). On alanine dipeptide, TICA’s oracle- k recovery (0.958) is clearly the best of the three, well above PCA (0.494) and VAE (0.391) – the largest gap between TICA and the other two methods observed on any system in this paper. Its selected- k recovery at the largest budget (0.249) remains far below this ceiling, the same selection-step bottleneck seen throughout. This is consistent with TICA’s construction: it is designed to find slow collective coordinates, and backbone dihedral transitions in a real molecule are exactly the kind of slow process TICA targets, whereas the Müller-Brown and Prinz potentials were not constructed to favor any particular method. The practical implication is that TICA’s relative advantage may be larger on real molecular systems than the synthetic benchmarks alone would suggest—a claim the synthetic systems could not have supported on their own, and that alanine dipeptide is well-suited to test given how it was built.

Replication on a second independent 100 ns seed. A second 100 ns trajectory was run under an identical protocol from a different random seed and swept over the budgets the two runs share (100 to 32,000 frames). This is a genuine physical replicate: it reran the full Langevin sampling process, not merely the subsampling and clustering randomness that the eight sweep seeds vary (Table 12).

The central claim replicates closely. No budget-inflation correlation moves by more than 0.02 between seeds, the

Table 10: TICA under coverage-matched subsampling on alanine dipeptide. Thinning to n frames from a 100,000-frame reference imposes a stride of $100,000/n$, so the effective physical lag shrinks as the budget grows. Projection quality is associated with the effective lag in this subsample sweep.

n_frames	100	1,000	8,000	16,000	64,000
effective lag	10 ns	1 ns	125 ps	62 ps	16 ps
oracle-k ARI	0.02	0.26	0.25	0.93	0.96

Thinning a trajectory rescales TICA's lag: recovery tracks τ_{eff} in the subsample sweep

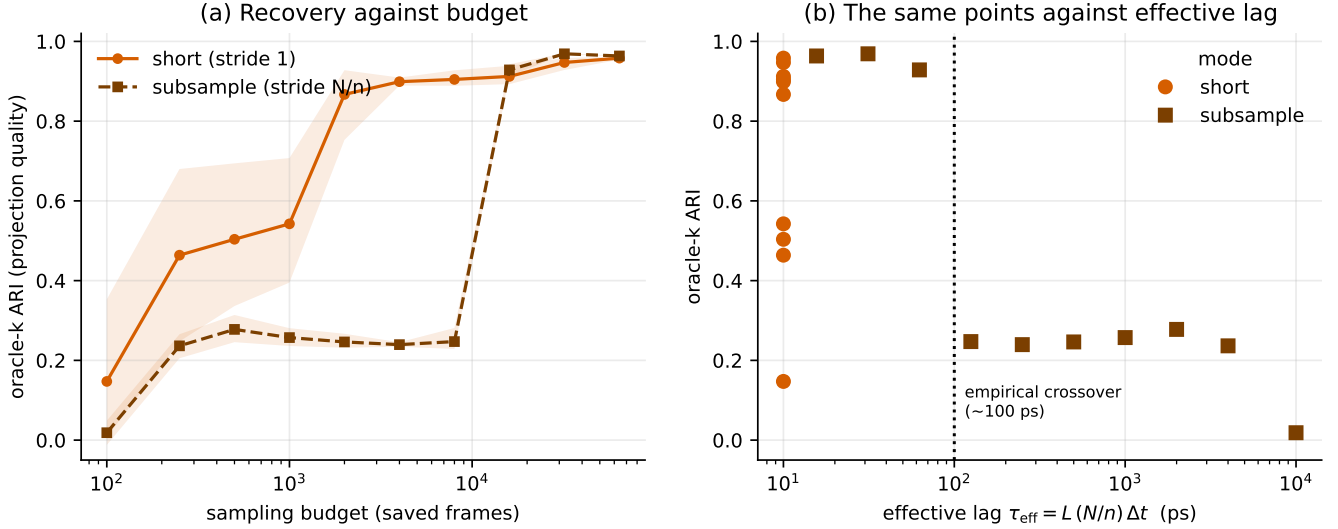


Figure 7: Thinning a trajectory rescales TICA’s lag. (a) Recovery against sampling budget for both modes. (b) The same points against the effective lag $\tau_{\text{eff}} = L(N/n)\Delta t$: every subsample point above the empirical 100 ps crossover sits near 0.25 and points below it jump to roughly 0.95. This association is descriptive, not a fitted transition timescale. The *short* points sit at a fixed 10 ps and still span 0.15 to 0.96, so their spread is sample size alone.

method ranking is unchanged with TICA’s oracle-k far above PCA’s and VAE’s on both, and basin visitation behaves identically (all three basins reached by $n=8,000$ under short trajectories and at every budget under subsampling).

One quantity does not replicate, and it is worth stating plainly rather than averaging away: **TICA shows no decline at all on seed 1 within the tested range.** Its recovery rises monotonically to the largest budget, where seed 0 had peaked at $n=16,000$ and fallen 27% by $n=32,000$. This is consistent with the crossover account in Section 6.3: fewer budget doublings remain after entry into the high-recovery regime. On seed 1 that threshold occurs one doubling later, at $n=32,000$ rather than $n=16,000$, leaving zero doublings in which selection-driven degradation could act. The oracle-k column shows the shift directly: at $n=16,000$ seed 0 has already reached 0.93 while seed 1 is only midway at 0.52, and both arrive at 0.96 by $n=32,000$. The seed-1 alanine case is therefore the same configuration as Prinz—warm-up consuming the entire sweep, zero doublings remaining, no decline observed—reached on a different system.

What this costs the paper is precision about *where* TICA’s crossover falls, not the BIC component-count trend: the recovery crossover is seed-dependent, so a single seed cannot

pin it down. The second seed is consistent with the proposed crossover account, but two physical replicates are insufficient to claim that the association has been prospectively confirmed.

8 Recommendations

1. **Report the sampling budget alongside any claimed metastable state count.** A state count without its budget is, by the evidence in this paper, an incompletely specified quantity.
2. **Do not report a state count from a single budget as a stable property of the system.** Where feasible, report how the count changes with budget (as in the complete method-system table), or explicitly justify why the budget used is believed to be past the point where this matters—which, per point 3, requires checking rather than assuming.
3. **Check the gap between recovery at the selected state count and at a plausible alternative before trusting the selected count, rather than assuming the selection step is reliable.** This paper’s oracle-k comparison is one way

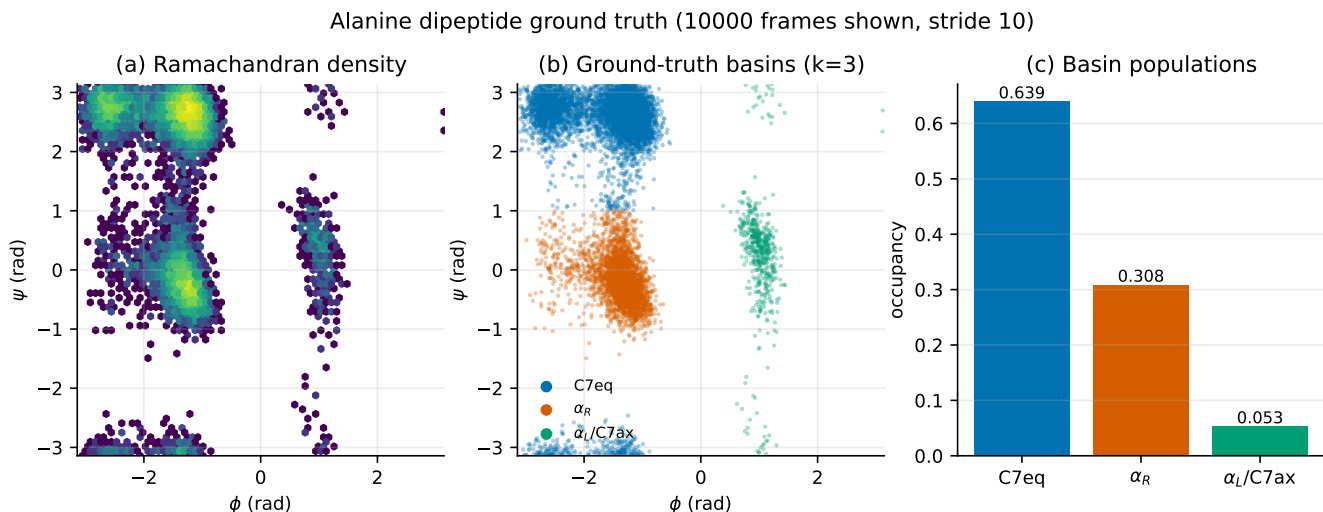


Figure 8: Alanine dipeptide structural reference, from the seed-1 trajectory: (a) Ramachandran density, (b) the standard three-region partition that supplies the basin labels, (c) basin occupancy. The partition is a chemical definition and is independent of every method under audit. The rarest basin holds 5.3% of seed-1 frames. These occupancies are not asserted to represent seed 0.

Table 11: Mean number of the three ground-truth basins visited by random contiguous windows of the 100 ns trajectory. Under coverage-matched subsampling all three are visited at every budget.

n_frames (short mode)	100	250	500	1000	2000	4000	8000+
mean basins visited (of 3)	2.0	2.1	2.1	2.3	2.6	2.8	3.0

to do this when ground truth is known; in practice a domain plausible-range check (does $k = 3$ versus $k = 8$ tell qualitatively different biological stories?) serves a similar function.

4. **Use ICL as a sensitivity analysis when fitting mixtures, not as a default answer.** In this grid ICL was less prone to ceiling saturation: it never reached the search ceiling in any of the twelve (system x mode x method) conditions tested here, where BIC did so in up to 44% of conditions and AIC in up to 84%. ICL’s own reported count should still be treated as an estimate to sanity-check, not a final answer or a kinetically validated state count.
5. **Where TICA is used, check for evidence of a low-recovery regime before trusting its output at a given budget.** TICA’s failure mode at small budgets (near-zero recovery, not merely inaccurate recovery) is distinct from PCA’s and VAE’s and is not fixed by any of the five selection criteria tested. It depends on budget, lag, the number of usable pairs, and the dynamical covariance targeted by that lag. The lag ablation in Section 6 shows generally lower recovery and a later crossover as lag increases, with small reversals that rule out a strict monotonic claim. A practical check even without ground truth is whether the projection changes substantially under a moderate change in lag; if it does, the estimate at the current lag and budget should not yet be trusted.

9 Limitations

Ordered by how much they constrain the paper’s claims, most binding first.

TICA’s high-recovery crossover budget is seed-dependent, so the size of its decline is not a fixed property of a system. The two 100 ns alanine seeds place that crossover one budget doubling apart, which is the difference between a 27% decline and none at all within the tested range (Section 7). The doublings-remaining mechanism accounts for both, and PCA and VAE replicate closely, but any single-seed estimate of *how far* TICA declines on a given system should be treated as one draw rather than a constant. Pinning the crossover down would need several seeds per system, which is not done here.

Two independent 100 ns seeds on alanine dipeptide, against twenty on each synthetic system. The real-system evidence is now genuinely replicated rather than single-seed, but two is still far short of the twenty independent Langevin runs behind each synthetic result, and the eight sweep seeds within each trajectory vary only subsampling, clustering initialization and VAE training rather than the underlying dynamics. Quantities that agreed closely across the two seeds are reported as replicated; the one that did not is reported as such in Section 7 rather than averaged.

An earlier 5 ns cross-seed comparison is retained only as provenance. Before the 100 ns runs, two independent 5 ns simulations were swept and compared, and the budget-inflation effect replicated across them. Both shared the basin-

Table 12: Two independent 100 ns alanine dipeptide seeds, compared over the budgets they share. Budget inflation, the method ranking and basin coverage all replicate. TICA’s relative decline does not: its high-recovery crossover occurs one budget doubling later on seed 1, leaving no room for a decline within the tested range. This is consistent with, but does not by itself validate, the crossover account.

quantity (budgets 100-32,000)	seed 0	seed 1
budget-inflation ρ , range over 6 mode x method cells	+0.87 to +0.97	+0.88 to +0.97
PCA relative decline, subsample	73%	73%
VAE relative decline, subsample	66%	63%
TICA relative decline, subsample	27%	0%
PCA oracle-k at n=32,000, subsample	0.473	0.537
TICA oracle-k at n=32,000, subsample	0.969	0.963
VAE oracle-k at n=32,000, subsample	0.385	0.481
first TICA oracle-k ARI ≥ 0.80	n=16,000	n=32,000

visitation deficit that motivated the move to 100 ns, so that agreement was never strong evidence about the properly-sampled result. It is now superseded outright by the 100 ns cross-seed comparison in Section 7 and is mentioned here only so the discarded pilot data in the repository is accounted for.

Silhouette and elbow-gap criteria are shown to be unreliable but not fully explained. Both return near-constant output regardless of budget, and this paper documents that fact and its consequence (an illusion of stability that happens to coincide with the correct answer on some systems and not others) without establishing why these particular criteria are so insensitive to this particular kind of data. This is noted as an open question rather than resolved.

The study counts geometric mixture components rather than validating kinetic metastability. No transition matrix, implied-timescale plateau, Chapman-Kolmogorov test, or PCCA-type coarse graining is applied. Consequently, the main result is about apparent-basin counts from a projection-plus-GMM pipeline. The synthetic reference basins support recovery calculations; alanine’s dihedral partition is only an independent structural reference.

All three systems tested have a small number of well-separated basins (3-4). Whether the effect’s magnitude changes for landscapes with many close or overlapping metastable states is untested and is a natural extension.

Alanine dipeptide, while a real molecule, is small and fast-mixing. The result should not be extrapolated to slower-folding or larger proteins without further validation; the paper’s claims about real systems are scoped to this system and are not evidence about protein folding timescales generally.

The alanine production protocol has no separate post-minimization equilibration stage. Production begins after energy minimization and velocity assignment. Random contiguous windows reduce dependence on the trajectory origin, but a future archive should include an explicit equilibration protocol and a sensitivity analysis that discards early production time.

Two selection-criterion claims rest on limited excursions. ICL’s zero ceiling-hit rate is a strong result but was tested only up to a search ceiling of $k_{\max} = 15$; whether ICL would eventually saturate at a still-larger budget is not established. Separately, ICL’s own accuracy (as opposed to its resistance to

runaway inflation) has not been separately validated against an independent ground truth beyond the ARI figures already reported.

TICA’s lag parameter was varied on one system only. The crossover account proposed in Section 6 is tested via a lag ablation (300 conditions, 10 seeds) rather than only inferred from the oracle-k contrast. The ablation supports lag sensitivity and a lag-dependent data requirement without isolating pair count as the cause. The lag parameter itself was swept on the Prinz potential only. Section 6.3 supplies a second, independent test on alanine dipeptide by varying the effective lag through the subsampling stride rather than through the parameter, and the crossover falls where the mechanism predicts; a direct parameter sweep on Müller-Brown and on alanine remains untested and would further strengthen the claim’s generality.

The embedding-dimension ablation is complete and confirms the main results are dimension-robust, with two secondary findings worth reporting. Re-running the core measurement at $n_{\text{components}}$ in $\{2, 3, 5\}$ on Prinz (540 conditions, 10 seeds) shows the budget-inflation effect essentially unchanged across dimension for all three methods (Spearman ρ stays within 0.86-0.98 with no consistent trend as dimension increases), and ICL’s zero ceiling-hit rate holds at all three dimensions (0 of 9 method-by-dimension combinations reach the cap), strengthening Recommendation 4 beyond the single dimension used elsewhere in the paper.

Two secondary effects emerged that were not anticipated. PCA’s oracle-k recovery improves monotonically with dimension (0.895 to 0.958 from $\text{dim} = 2$ to $\text{dim} = 5$ at the largest budget), consistent with a purely linear projection retaining more information as more output dimensions are allowed; TICA’s and VAE’s oracle-k recovery is roughly flat or mildly declines over the same range, consistent with both already capturing the dynamically relevant directions at two dimensions, leaving additional dimensions to add unneeded capacity (VAE) or extra room for the selection step to overfit (TICA). Because selected-k recovery barely moves with dimension for any method, PCA’s oracle-versus- selected gap widens as dimension increases—a cleaner illustration of the paper’s localization claim than the two-dimensional case alone, since it shows the selection step’s relative cost growing precisely as

the projection itself improves. This ablation was run on Prinz only; whether the same pattern holds on Müller-Brown and alanine dipeptide is untested.

Data availability

All result files underlying every number and figure in this paper are in the accompanying repository at <https://github.com/imjbassi/sampling-audit>, under `results/`: `sweep.csv` and `sweep_prinz.csv` (960 conditions each), `alanine_sweep_100ns.csv` (480 conditions), `alanine_sweep_100ns_seed1.csv` (432 conditions), `tica_lag_ablation.csv` (300 conditions) and `dim_ablation.csv` (540 conditions), and `iid_equilibrium_control.csv` (160 conditions). The superseded 5 ns alanine pilot sweeps are retained in the same directory for provenance and are identified as such in `results/NOTE.md`. The alanine starting structure, complete simulation protocol, random-seed controls and derived result tables are included. The exact production trajectories are not currently archived in the repository; rerunning molecular dynamics reproduces the protocol but is not expected to be bitwise identical across hardware and software stacks. This is a remaining FAIR-data limitation and should be resolved with a versioned trajectory archive and DOI before journal submission.

Code availability

All code is in the same repository under the MIT License. `run_all.sh` writes a fresh, non-resuming reproduction to a separate output directory and includes the synthetic sweeps and registered ablations; `src/alanine.py` runs the molecular dynamics and its sweep; `src/figures.py` and `src/figures_alanine.py` regenerate every figure from the result CSVs.

Competing interests

The author declares no competing interests.

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Author contributions

Jaiveer Bassi: conceptualization, methodology, software, formal analysis, investigation, visualization, writing—original draft, and writing—review and editing.

Use of AI tools

OpenAI ChatGPT and Codex were used to assist with manuscript editing, consistency checking, software review, statistical cross-checking, and LaTeX build troubleshooting. The author independently reviewed and takes responsibility for the scientific reasoning, computations, references, and final text.

References

- [1] Andrea Amadei, Antonius B. M. Linssen, and Herman J. C. Berendsen. Essential dynamics of proteins. *Proteins: Structure, Function, and Bioinformatics*, 17(4):412–425, December 1993.
- [2] Guillermo Pérez-Hernández, Fabian Paul, Toni Giorgino, Gianni De Fabritiis, and Frank Noé. Identification of slow molecular order parameters for Markov model construction. *The Journal of Chemical Physics*, 139(1):015102, July 2013.
- [3] Christian R. Schwantes and Vijay S. Pande. Improvements in Markov state model construction reveal many non-native interactions in the folding of NTL9. *Journal of Chemical Theory and Computation*, 9(4):2000–2009, March 2013.
- [4] Carlos X. Hernández, Hannah K. Wayment-Steele, Mohammad M. Sultan, Brooke E. Husic, and Vijay S. Pande. Variational encoding of complex dynamics. *Physical Review E*, 97(6):062412, June 2018.
- [5] Christoph Wehmeyer and Frank Noé. Time-lagged autoencoders: Deep learning of slow collective variables for molecular kinetics. *The Journal of Chemical Physics*, 148(24):241703, March 2018.
- [6] João Marcelo Lamim Ribeiro, Pablo Bravo, Yihang Wang, and Pratyush Tiwary. Reweighted autoencoded variational Bayes for enhanced sampling (RAVE). *The Journal of Chemical Physics*, 149(7):072301, May 2018.
- [7] Brooke E. Husic and Vijay S. Pande. Markov state models: From an art to a science. *Journal of the American Chemical Society*, 140(7):2386–2396, February 2018.
- [8] John D Chodera and Frank Noé. Markov state models of biomolecular conformational dynamics. *Current Opinion in Structural Biology*, 25:135–144, April 2014.
- [9] Martin K. Scherer, Benjamin Trendelkamp-Schroer, Fabian Paul, Guillermo Pérez-Hernández, Moritz Hoffmann, Nuria Plattner, Christoph Wehmeyer, Jan-Hendrik Prinz, and Frank Noé. PyEMMA 2: A software package for estimation, validation, and analysis of Markov models. *Journal of Chemical Theory and Computation*, 11(11):5525–5542, October 2015.

- [10] Aldo Glielmo, Brooke E. Husic, Alex Rodriguez, Cecilia Clementi, Frank Noé, and Alessandro Laio. Unsupervised learning methods for molecular simulation data. *Chemical Reviews*, 121(16):9722–9758, May 2021.
- [11] Xinyan Wu, Zhiteng Zhang, Donghui Shao, Xuyang Liu, Jingya Xing, Haohao Fu, Wensheng Cai, and Xueguang Shao. A tutorial on dimensionality reduction and clustering for molecular dynamics trajectories: From linear algorithms to deep learning. *The Journal of Physical Chemistry B*, 130(30):7499–7511, 2026.
- [12] Alan Grossfield and Daniel M. Zuckerman. *Chapter 2 Quantifying Uncertainty and Sampling Quality in Biomolecular Simulations*, pages 23–48. Elsevier, 2009.
- [13] Gideon Schwarz. Estimating the dimension of a model. *The Annals of Statistics*, 6(2):461–464, March 1978.
- [14] Christine Keribin. Consistent estimation of the order of mixture models. *Sankhyā: The Indian Journal of Statistics, Series A*, 62(1):49–66, 2000.
- [15] Mathias Drton and Martyn Plummer. A Bayesian information criterion for singular models. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 79(2):323–380, February 2017.
- [16] Klaus Müller and Leo D. Brown. Location of saddle points and minimum energy paths by a constrained simplex optimization procedure. *Theoretica Chimica Acta*, 53(1):75–93, 1979.
- [17] Jan-Hendrik Prinz, Hao Wu, Marco Sarich, Bettina Keller, Martin Senne, Martin Held, John D. Chodera, Christof Schütte, and Frank Noé. Markov models of molecular kinetics: Generation and validation. *The Journal of Chemical Physics*, 134(17):174105, May 2011.
- [18] Peter Eastman, Jason Swails, John D. Chodera, Robert T. McGibbon, Yutong Zhao, Kyle A. Beauchamp, Lee-Ping Wang, Andrew C. Simmonett, Matthew P. Harrigan, Chaya D. Stern, Rafal P. Wiewiora, Bernard R. Brooks, and Vijay S. Pande. OpenMM 7: Rapid development of high performance algorithms for molecular dynamics. *PLOS Computational Biology*, 13(7):e1005659, July 2017.
- [19] James A. Maier, Carmenza Martinez, Koushik Kasavajhala, Lauren Wickstrom, Kevin E. Hauser, and Carlos Simmerling. ff14SB: Improving the accuracy of protein side chain and backbone parameters from ff99SB. *Journal of Chemical Theory and Computation*, 11(8):3696–3713, July 2015.
- [20] Hai Nguyen, Daniel R. Roe, and Carlos Simmerling. Improved generalized Born solvent model parameters for protein simulations. *Journal of Chemical Theory and Computation*, 9(4):2020–2034, March 2013.
- [21] Diederik P. Kingma and Max Welling. Auto-encoding variational Bayes. In *2nd International Conference on Learning Representations (ICLR 2014), Conference Track Proceedings*, Banff, AB, Canada, 2014.
- [22] Laurens van der Maaten and Geoffrey Hinton. Visualizing data using t-SNE. *Journal of Machine Learning Research*, 9(86):2579–2605, 2008.
- [23] Martin Wattenberg, Fernanda Viégas, and Ian Johnson. How to use t-SNE effectively. *Distill*, 1(10), October 2016.
- [24] H. Akaike. A new look at the statistical model identification. *IEEE Transactions on Automatic Control*, 19(6):716–723, December 1974.
- [25] C. Biernacki, G. Celeux, and G. Govaert. Assessing a mixture model for clustering with the integrated completed likelihood. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 22(7):719–725, July 2000.
- [26] Peter J. Rousseeuw. Silhouettes: A graphical aid to the interpretation and validation of cluster analysis. *Journal of Computational and Applied Mathematics*, 20:53–65, November 1987.
- [27] Lawrence Hubert and Phipps Arabie. Comparing partitions. *Journal of Classification*, 2(1):193–218, December 1985.
- [28] Zhijun Zhang, Xinzijian Liu, Kangyu Yan, Mark E. Tuckerman, and Jian Liu. Unified efficient thermostat scheme for the canonical ensemble with holonomic or isokinetic constraints via molecular dynamics. *The Journal of Physical Chemistry A*, 123(28):6056–6079, May 2019.
- [29] Robert T. McGibbon, Kyle A. Beauchamp, Matthew P. Harrigan, Christoph Klein, Jason M. Swails, Carlos X. Hernández, Christian R. Schwantes, Lee-Ping Wang, Thomas J. Lane, and Vijay S. Pande. MDTraj: A modern open library for the analysis of molecular dynamics trajectories. *Biophysical Journal*, 109(8):1528–1532, October 2015.